

Elementary Operations on Quasi-Toeplitz Matrices

PCCA6

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Introduction

Definitions

Toeplitz M.
Displacement M.
Reconstruction
Multiplication
Addition

Compression

Inversion

Benchmarking

Toeplitz Matrix A_1

$$A_1 = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 \end{pmatrix}$$

Where do we see these structured matrices?

- Sylvester matrix
- Diagonal gradients
- Quantum channel calculations in classical quantum simulations
- Digital Signal Processing
- ...

Definition

A matrix $A \in \mathbb{K}^{n \times m}$ is **Toeplitz** if $A_{i,j} = A_{i+1,j+1}$ for all valid i, j .

$$A = \begin{pmatrix} a_0 & a_{-1} & \cdots & \cdots & a_{-(m-1)} \\ a_1 & a_0 & \cdots & \cdots & a_{-(m-2)} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ a_{n-1} & \cdots & a_1 & a_0 & a_{m-n} \end{pmatrix}$$

Examples of Toeplitz Matrices

$$A_1 = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 \end{pmatrix} \quad A_2 = \begin{pmatrix} 120 & 432 & 584 & 394 & 159 \\ 312 & 120 & 432 & 584 & 394 \\ 985 & 312 & 120 & 432 & 584 \end{pmatrix}$$

Displacement Operator

$$\nabla A = A - ZAZ^T = GH^T$$

For Toeplitz: $\text{rank}(\nabla A) \leq 2$

Shift Matrix

$$Z = \begin{pmatrix} 0 & & & \\ 1 & & & \\ & \ddots & & \\ & & 1 & 0 \end{pmatrix} \quad Z^T = \begin{pmatrix} 0 & 1 & & \\ & & \ddots & \\ & & & 1 \\ & & & 0 \end{pmatrix}$$

Displacement Operator

$$\nabla A = A - ZAZ^T = GH^T$$

For Toeplitz: $\text{rank}(\nabla A) \leq 2$

Displacement $\nabla A = A - ZAZ^T$

$$A_1 = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 \end{pmatrix} \quad \nabla A_1 = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 \end{pmatrix}$$

Only first row and column stays. $\text{rank} = 2$.

Displacement Matrix A

$$\nabla A_1 = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 \end{pmatrix}$$

Generators G and H

$$G = \begin{pmatrix} 4 & 0 \\ 5 & 1 \\ 6 & 0 \\ 7 & 0 \end{pmatrix} \quad H^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 2 & 1 \end{pmatrix}$$

Reconstruction

$$A = \sum_{k=1}^{\alpha} L(g_k) U(h_k)$$

Defining L and U are Lower and Upper matrices respectively:

$$L(g_k) = \begin{bmatrix} g_1 & 0 & 0 & 0 \\ g_2 & g_1 & 0 & 0 \\ g_3 & g_2 & g_1 & 0 \\ g_4 & g_3 & g_2 & g_1 \end{bmatrix} \quad U(h_k) = \begin{bmatrix} h_1 & h_2 & h_3 & h_4 \\ 0 & h_1 & h_2 & h_3 \\ 0 & 0 & h_1 & h_2 \\ 0 & 0 & 0 & h_1 \end{bmatrix}$$

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$$A = L(g^{(1)})U(h^{(1)}) + L(g^{(2)})U(h^{(2)})$$

$$L(g^{(1)})U(h^{(1)}) = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 5 & 4 & 0 & 0 \\ 6 & 5 & 4 & 0 \\ 7 & 6 & 5 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 5 & 4 & 0 & 0 \\ 6 & 5 & 4 & 0 \\ 7 & 6 & 5 & 4 \end{pmatrix}$$

$$L(g^{(2)})U(h^{(2)}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 3 & 2 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 2 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Reconstruction

$$A = \sum_{k=1}^{\alpha} L(g_k) U(h_k)$$

Complexities

Naive: $O(n^3)$

polynome products : $O(n^2)$

Multiplication of Toeplitz-like Matrices

Generator Multiplication with Vectors

$$A = \sum_{k=1}^{\alpha} L(g_k) U(h_k) V$$

Complexities

Naive: $O(n^3)$

polynome products : $O(\alpha M(n))$

Multiplication of Toeplitz-like Matrices

Generator Multiplication

$$\begin{aligned} A &\approx (T, U), & B &\approx (G, H) \\ W &= ZAZ^t G, & V &= B^t U \\ \text{LastCol} &= (0 \cdots 0, 1), & a &= ZALastCol, & b &= ZB^t \text{LastCol} \\ C &\approx (T|W|a, V|H|b) \end{aligned}$$

Complexities

Naive: $O(n^3)$

Concatenate : $O(\alpha^2 M(n))$

Generator Addition

$$A \simeq (T, U), B \simeq (G, H) \Rightarrow A + B \simeq (T|G, U|H)$$

Complexities

Naive: $O(n^2)$

Concatenate : $O(\alpha n)$

$$C = 2A \quad \text{with} \quad A_1 = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 \\ 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 \end{pmatrix}, G = \begin{pmatrix} 4 & 0 \\ 5 & 1 \\ 6 & 0 \\ 7 & 0 \end{pmatrix}, H^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 2 & 1 \end{pmatrix}$$

$$T = \begin{pmatrix} 4 & 0 & 4 & 0 \\ 5 & 1 & 5 & 1 \\ 6 & 0 & 6 & 0 \\ 7 & 0 & 7 & 0 \end{pmatrix}, U = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 3 & 0 & 3 & 0 \\ 2 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

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$$C = L(g^{(1)})U(h^{(1)}) + L(g^{(2)})U(h^{(2)}) + L(g^{(3)})U(h^{(3)})L(g^{(4)})U(h^{(4)})$$

$$\begin{pmatrix} 4 & 0 & 0 & 0 \\ 5 & 4 & 0 & 0 \\ 6 & 5 & 4 & 0 \\ 7 & 6 & 5 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 3 & 2 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 4 & 0 & 0 & 0 \\ 5 & 4 & 0 & 0 \\ 6 & 5 & 4 & 0 \\ 7 & 6 & 5 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 3 & 2 & 1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 6 & 4 & 2 \\ 10 & 8 & 6 & 4 \\ 12 & 10 & 8 & 6 \\ 14 & 12 & 10 & 8 \end{pmatrix}$$

Operations like addition and multiplication forcefully increase the rang of the generators.

$$G = \begin{pmatrix} 4 & \dots \\ 5 & \dots \\ 6 & \dots \\ 7 & \dots \end{pmatrix} \quad H^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

Definition

Matrices $G \in \mathbb{R}^{n \times \alpha}$ and $H \in \mathbb{R}^{n \times \alpha}$ represent a matrix $\nabla A = GH^T$.

Goal: Find two matrices $G' \in \mathbb{R}^{n \times r}$ and $H' \in \mathbb{R}^{n \times r}$ such that:

$$G'H'^T = GH^T$$

Algorithm 1 Generator Compression

```

1: function GR_MAT_GENERATOR_COMPRESS_AUX( $G, H$ )
2:    $P, L, U \leftarrow \text{LU\_Decomposition}(G^T)$ 
3:    $r \leftarrow$  Number of non-zero rows in  $U$ 
4:   if  $r < 1$  then
5:     return Generators of a 0 matrix
6:   end if
7:    $G_D \leftarrow (U_{[0:r-1],*})^T$ 
8:    $H_D \leftarrow HP^T \times L_{*,[0:r-1]}$ 
9:   return  $G_D, H_D$ 
10: end function

```

$\triangleright PG^T = LU$
 $\triangleright \text{rank}$

Giving us a complexity of $O(\alpha^2 n)$

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Inverting a Toeplitz matrix represented by generators allows us to efficiently solve systems. But can also be used in:

- Image unblurring and de-echo.
- Speech compression and financial forecasting.

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Inverting a Toeplitz matrix represented by generators allows us to efficiently solve systems.

We can apply a divide-and-conquer algorithm, aiming to reduce the complexity from $O(n^3)$ to $O(n \log^2 n)$.

Algorithm 2 Strassen-type Inversion for Generators: $\text{STRASSENINV}(G_A, H_A)$

```

1: if  $n = 1$  then return scalar inverse  $(G_D, H_D)$ 
2: end if
3:  $G_a, H_a, \dots, G_d, H_d \leftarrow \text{SPLIT}(G_A, H_A)$ 
4:  $G_e, H_e \leftarrow \text{STRASSENINV}(G_a, H_a)$ 
5:  $G_{ce}, H_{ce} \leftarrow \mathcal{C}(G_c, H_c \otimes G_e, H_e)$ 
6:  $G_{eb}, H_{eb} \leftarrow \mathcal{C}(G_e, H_e \otimes G_b, H_b)$ 
7:  $G_{ceb}, H_{ceb} \leftarrow \mathcal{C}(G_c, H_c \otimes G_{eb}, H_{eb} \times -1)$ 
8:  $G_S, H_S \leftarrow \mathcal{C}(G_d, H_d \oplus G_{ceb}, H_{ceb})$ 
9:  $G_t, H_t \leftarrow \text{STRASSENINV}(G_S, H_S)$ 
10:  $G_{ebt}, H_{ebt} \leftarrow \mathcal{C}(G_{eb}, H_{eb} \otimes G_t, H_t)$ 
11:  $G_{tce}, H_{tce} \leftarrow \mathcal{C}(G_t, H_t \otimes G_{ce}, H_{ce})$ 
12:  $G_{ebtce}, H_{ebtce} \leftarrow \mathcal{C}(G_{ebt}, H_{ebt} \otimes G_{ce}, H_{ce})$ 
13:  $G_x, H_x \leftarrow \mathcal{C}(G_e, H_e \oplus G_{ebtce}, H_{ebtce})$ 
14:  $G_y, H_y \leftarrow \text{NEGATE}(G_{ebt}, H_{ebt}); \quad G_z, H_z \leftarrow \text{NEGATE}(G_{tce}, H_{tce})$ 
15:  $G_{Inv}, H_{Inv} \leftarrow \text{PACK}(G_x, H_x, G_y, H_y, G_z, H_z, G_t, H_t)$ 
16: return  $\mathcal{C}(G_{Inv}, H_{Inv})$ 

```

▷ Step 1: Base Case
 ▷ Step 2: Quadrants
 ▷ Step 3: Recursion 1 ($e := a^{-1}$)
 ▷ Step 4: Schur Complement

 ▷ $S := d - ceb$
 ▷ Step 5: Recursion 2 ($t := S^{-1}$)
 ▷ Step 6: Strassen Assembly

 ▷ $x := e + ebtce$

 ▷ Step 7: Pack

Notation: $\mathcal{C}(\cdot)$ = Compress, $\otimes$ = Multiply Generators, $\oplus$ = Add Generators

Algorithm 3 Strassen-type Inversion for Generators: $\text{STRASSENINV}(G_A, H_A)$

```

1: if  $n = 1$  then return scalar inverse  $(G_D, H_D)$ 
2: end if
3:  $G_a, H_a, \dots, G_d, H_d \leftarrow \text{SPLIT}(G_A, H_A)$ 
4:  $G_e, H_e \leftarrow \text{STRASSENINV}(G_a, H_a)$ 
5:  $G_{ce}, H_{ce} \leftarrow \mathcal{C}(G_c, H_c \otimes G_e, H_e)$ 
6:  $G_{eb}, H_{eb} \leftarrow \mathcal{C}(G_e, H_e \otimes G_b, H_b)$ 
7:  $G_{ceb}, H_{ceb} \leftarrow \mathcal{C}(G_c, H_c \otimes G_{eb}, H_{eb} \times -1)$ 
8:  $G_S, H_S \leftarrow \mathcal{C}(G_d, H_d \oplus G_{ceb}, H_{ceb})$ 
9:  $G_t, H_t \leftarrow \text{STRASSENINV}(G_S, H_S)$ 
10:  $G_{ebt}, H_{ebt} \leftarrow \mathcal{C}(G_{eb}, H_{eb} \otimes G_t, H_t)$ 
11:  $G_{tce}, H_{tce} \leftarrow \mathcal{C}(G_t, H_t \otimes G_{ce}, H_{ce})$ 
12:  $G_{ebtce}, H_{ebtce} \leftarrow \mathcal{C}(G_{ebt}, H_{ebt} \otimes G_{ce}, H_{ce})$ 
13:  $G_x, H_x \leftarrow \mathcal{C}(G_e, H_e \oplus G_{ebtce}, H_{ebtce})$ 
14:  $G_y, H_y \leftarrow \text{NEGATE}(G_{ebt}, H_{ebt}); \quad G_z, H_z \leftarrow \text{NEGATE}(G_{tce}, H_{tce})$ 
15:  $G_{Inv}, H_{Inv} \leftarrow \text{Pack}(G_x, H_x, G_y, H_y, G_z, H_z, G_t, H_t)$ 
16: return  $\mathcal{C}(G_{Inv}, H_{Inv})$ 

```

▷ **Step 1: Base Case**
 ▷ **Step 2: Quadrants**
 ▷ **Step 3: Recursion 1** ($e := a^{-1}$)
 ▷ **Step 4: Schur Complement**

 ▷ $S := d - ceb$
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 ▷ **Step 7: Pack**

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Live Benchmarking

Thank you